

Slingshot Controller

SUMMER TRAINING REPORT

Submitted in the partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

ELECTRICAL AND ELECTRONICS ENGINEERING



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DECLARATION

I hereby declare that the project work/training entitled as **“Slingshot Controller”** is an authentic record of my own work carried out at **EEE, UIET, Panjab University, Chandigarh** as requirements of 2nd semester Summer Training for the award of the degree of B.E. at University Institute of Engineering & Technology, Panjab University, Chandigarh under the guidance of **Mr. Kuldeep Singh** from 27-05-2024 to 11-06-2024 (15 Days).

Date :

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Acknowledgement

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We would like to express our sincere thanks to all of our DEPARTMENT FACULTY MEMBERS AND NON-TEACHING STAFF MEMBERS of the Department of UIET who gave many suggestions from time to time that made our project work better and well finished. Last but not the least we would like to thank our parents and friends for the supports, concerns and prayers, which were a major factor in the completion of this project.

Introduction

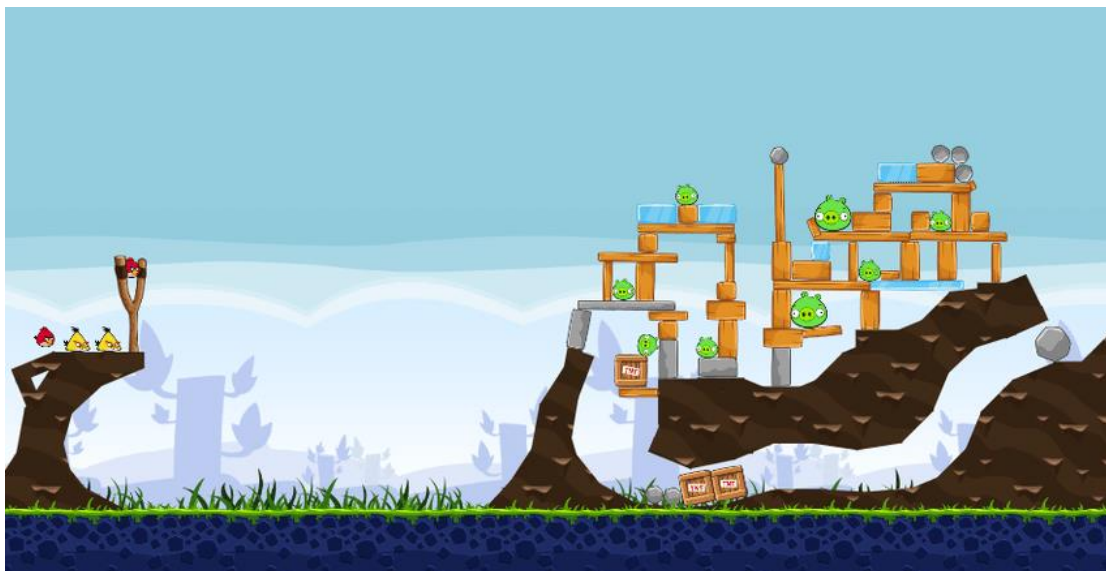
This project aims to create a DIY slingshot that can be used to play the popular Angry Birds game on a computer. The slingshot is made using simple materials and controlled by an Arduino, which is a small programmable computer. In this project, we will build a slingshot frame, connect it to Arduino and use a potentiometer to measure the angle at which the slingshot is pulled back. A button will be used to trigger the launch and a Flex sensor which will work as the Force component of the slingshot. We will then program the Arduino to send data to a computer, where a Mouse library will be used to simulate mouse movements to control the Angry Birds game. The combination of hardware and software allows us to interact with the game in a fun and engaging way. By the end, we will have a functional slingshot that lets us play Angry Birds using physical actions, bringing a new level of interaction to the game. This project combines elements of electronics, programming and mechanical design, providing a comprehensive learning experience in creating interactive devices.

Software and Hardware Requirement

SOFTWARE AND HARDWARE REQUIREMENTS:

SOFTWARE:

- ARDUINO IDE
- PROCESSING IDE
- ANGRY BIRDS GAME ON COMPUTER



Hardware:

- Arduino (Any Version)
- Potentiometer
- Flex Sensor
- 320 ohm Resistor
- Connecting Wires
- Breadboard

- **ARDUINO IDE**

The Arduino integrated development environment (ide) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. It is used to write and upload programs to Arduino compatible boards



Angry Birds Game on Computer

HARDWARE:

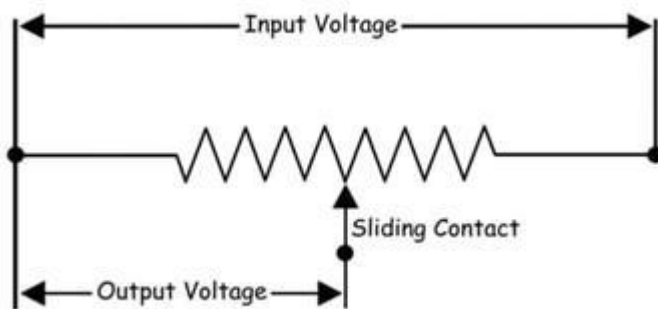
-ARDUINO (ANY VERSION)

Arduino is an open-source prototyping platform used for building electronics projects. It consists of both a physical programmable circuit board and a software, or ide (integrated development environment) that runs on your computer, where you can write and upload the computer code to the physical board.

-POTENTIOMETER

WHAT IS A POTENTIOMETER?

A potentiometer (also known as a **pot** or **potmeter**) is defined as a 3 terminal variable resistor in which the resistance is manually varied to control the flow of electric current. A potentiometer acts as an adjustable voltage divider.

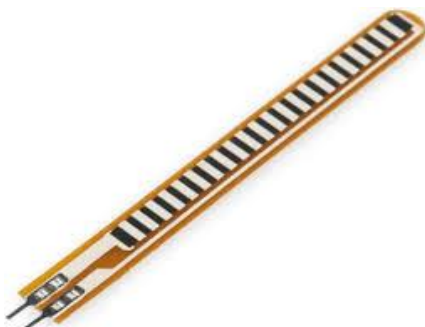


- Potentiometers work by varying the position of a **sliding contact** across a uniform resistance.

• FLEX SENSOR

WHAT IS A FLEX SENSOR?

Flex sensor is a kind of sensor which is used to measure the amount of deflection otherwise bending. The designing of this sensor can be done by using materials like plastic and carbon. The carbon surface is arranged on a plastic strip as this strip is turned aside then the sensor's resistance will be changed. Thus, it is also named a bend sensor. As its varying resistance can be directly proportional to the quantity of turn thus it can also be employed like a goniometer.



Concept Behind

The **Arduino Board** reads the values from the **Potentiometer and Flex Sensor** and Transmits them to the Laptop/PC through USB COM port via normal *analogRead()* function.

We then set an initial position and a final position and use the change in position to move the mouse using Mouse Core Library of Arduino. We use *Mouse.move()* method for movement of the mouse.

We then use a push button which works as trigger to fire the birds using *Mouse.press()* function.

The whole system combines and gives us a working slingshot launcher which has a flex sensor for X axis of mouse, a potentiometer for Y axis and a Push button for left click of mouse.

Working of Project

We use a Flex sensor for the X axis movement of the mouse (which acts as force for the slingshot) and the Potentiometer for the Y axis (which acts as the change in angle of the slingshot). The project uses the core Mouse Library to convert Analog signals from the sensors to convert them into movements from the mouse. The project uses the change in readings in the sensors and convert the change into movement of position from the mouse. We use a push button and use digital signals to use it as a mouse left click and trigger the launch of the bird. The code further contains an average filter for the smoothing of Y axis. The X axis only works when there is a change of 15 units in the flex sensor and moves 30 blocks to the left of the screen when the condition is satisfied. The project is then fitted into a self-made frame for ease of operation. After the appropriate connections are made, the model is ready and works as a launcher for the Angry Birds game. The launcher contains a trigger connected to a rubber band which pulls it back. The trigger flexes the flex sensor after which the angle of the launcher can be adjusted, The angular movement of the launcher rotates the potentiometer connected to it providing angular movement in the game. After which the push button can be used to trigger the launch of the bird.

Circuit Diagram And Photos

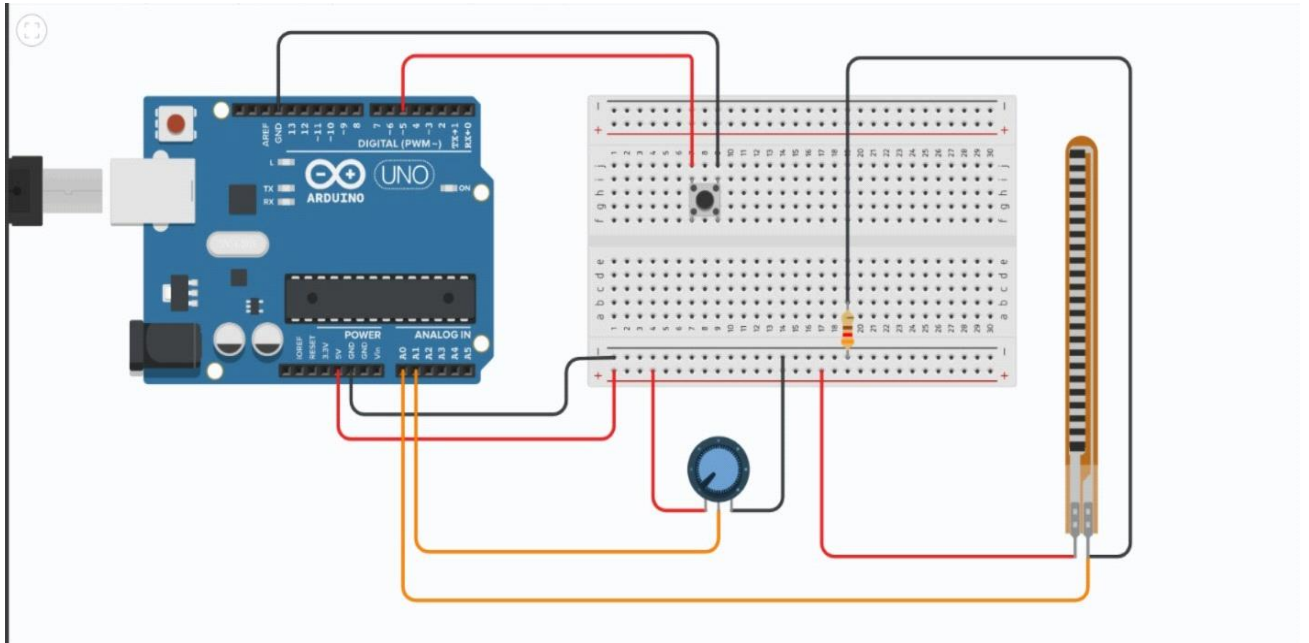


Fig 1

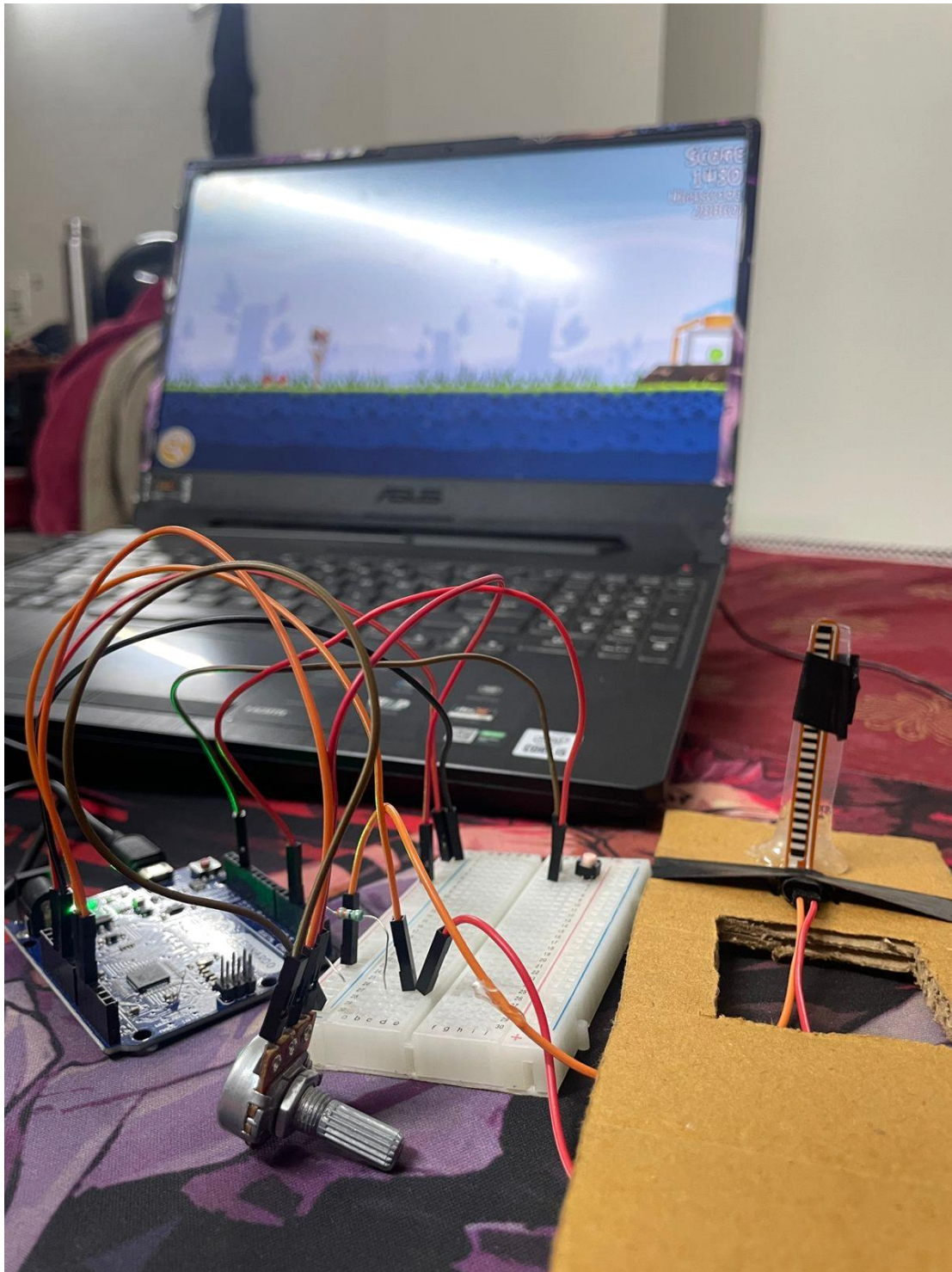


Fig 2

CODES

Arduino Code for Slingshot Controller

```
#include <Mouse.h>

int xsPull = 0; //start position of X
int xePull = 0; //end position of X
int ysPull = 0; //start position of Y
int yePull = 0; //end position of Y
int pushButton = 4; //push button for holding the bird
int range = 15;
int responseDelay = 1;
int buttonState = 0;
int xDistance = -50;

//Initailization for Smoothing of Y
//Using a simple moving Average Filter
const int numReadings = 10;
int readings[numReadings]; // the readings from the analog input
int index = 0; // the index of the current reading
int total = 0; // the running total
int average = 0; // the average

void setup() {

    Serial.begin(9600);
```

```

Serial.begin(9600);
Mouse.begin();
pinMode(pushButton, INPUT_PULLUP);

// Smoothing initialisation
for (int thisReading = 0; thisReading < numReadings; thisReading++)
  readings[thisReading] = 0;
}

void loop() {

  int xValue = analogRead(A0);
  if(xValue >= 2 || xValue <= 13){
    xsPull = map(xValue, 13,2,100,0);
  }

  ysPull = analogRead(A1);

  xePull=xsPull;
  delay(100);

  xValue= analogRead(A0); // Read value of X again
  if(xValue >= 2 || xValue <= 13){
    xsPull = map(xValue, 13,2,100,0);

  if(xValue >= 2 || xValue <= 13){
    xsPull = map(xValue, 13,2,100,0);
  }

  int xPull = xePull-xsPull; // Delta X

  yePull=ysPull;

  ysPull = analogRead(A1);

  int yPull = yePull - ysPull; // Delta Y

  buttonState= digitalRead(pushButton);

  if (buttonState == LOW) {
    Mouse.press(MOUSE_LEFT);
  }

  else {
    Mouse.release();
  }
}

```

```

if (xPull<=-15|| xPull>=15){
    Mouse.move(xDistance,0,0);
}

//Subtracting last reading from total
total= total - readings[index];
// read from the sensor:
readings[index] = yPull;
// add the reading to the total:
total= total + readings[index];
// advance to the next position in the array:
index = index + 1;

// if we're at the end of the array...
if (index >= numReadings)
    // ...wrap around to the beginning:
    index = 0;

// calculate the average:
average = total / numReadings;
// if we're at the end of the array...
if (index >= numReadings)
    // ...wrap around to the beginning:
    index = 0;

// calculate the average:
average = total / numReadings;

delay(1);
int yDistance = (average)/2;

if (yDistance != 0) {

    Mouse.move(0, -yDistance, 0);

}

// a delay so the mouse doesn't move too fast:
delay(responseDelay);

}

```

References

- [Arduino Reference - Arduino Reference](#)
- [Mouse - Arduino Reference](#)
- [Doing multiple timed things with Arduino: Unleash the millis\(\)! - YouTube](#)
- [Arduino MASTERCLASS | Full Programming Workshop in 90 Minutes! \(youtube.com\)](#)
- [AutoHotkey 4: MouseClick at specific position \(youtube.com\)](#)